Leveraging Domain Adaptation for Accurate Machine Learning Predictions of New Halide Perovskites

Dipannoy Das Gupta,[†,‡] Zachary J. L. Bare,[†] John T. Barber ,[†] Teerachote Pakornchote,[†] Santosh Adhikari,[†] Suxuen Yew,[¶] Brian DeCost,[§] Qi Zhang,[‡] Charles Musgrave,[¶] and Christopher Sutton[*,†]

†*Department of Chemistry and Biochemistry, University of South Carolina, Columbia, SC 29208, United States*

‡*Department of Computer Science and Engineering, University of South Carolina, Columbia, SC 29208, United States*

¶*Department of Chemical and Biological Engineering, University of Colorado, Boulder, CO 80303, United States*

§*National Institute of Standards and Technology, Gaithersburg, MD 20899, United States*

E-mail: cs113@mailbox.sc.edu

Abstract

We combine graph neural networks (GNN) with an inexpensive and reliable structure generation approach based on the bond-valence method (BVM) to train accurate machine learning models for screening 222,960 halide perovskites using statistical estimates of the DFT/PBE formation energy (ΔH_f^{PBE}), and the PBE (E_g^{PBE}) and HSE (E_g^{HSE}) band gaps. The GNNs were fined tuned using domain adaptation (DA) from a source model, which yields a factor of 1.6 times improvement in ΔH_f^{PBE} and 1.2 -

1.35 times improvement in E_g^{HSE} compared to direct training (i.e., without DA). As far as we are aware, this is the first demonstration of DA for GNNs in materials science. Moreover, after three active learning iterations – resulting in a total of 3593 training samples for ΔH_f – the best ML model produced mean absolute error (MAE) of 38 meV/atom relative to ΔH_f^{PBE} for totally unseen materials. The ML model for predicting E_g^{HSE} achieved an MAE of 0.18 eV. Using these two ML models, 48 compounds were identified out of 222,960 candidates as both stable and that have an HSE band gap that is relevant for photovoltaic applications. For this subset, only 8 have been reported to date, indicating that 40 compounds remain unexplored to the best of our knowledge and therefore offer opportunities for potential experimental examination.

Introduction

Halide perovskites are a broadly used class of materials for a variety of applications, including light-emitting diodes (LEDs) and photocatalysis that are particularly promising as light-adsorbing materials for photovoltaics (PV).[1–4] One reason for the wide applicability of perovskites is their extensive chemical space,[5] which enables modulation of their optical, electronic, and thermodynamic properties by changing their compositions. Although the enormous chemical space opens up opportunities for fine-tuning the properties of potential materials, it simultaneously poses a significant challenge for effectively screening materials.

Recent improvements in computing power allow for the use of quantum mechanical calculations, e.g., density functional theory (DFT), for high-throughput screening of materials. Specifically for screening perovskites, Kar et al. and Schmidt et al. have examined ABX_3 compounds.[6–8] However, AA'BB'X_6 perovskites are less explored because their unit cells are at least twice as large as those of ABX_3 perovskites, and thus, approximately 8 times more computationally expensive when assuming cubic scaling (N^3) of DFT. Alternatively, machine learning (ML) can be used to accurately predict DFT properties of atomistic systems at orders of magnitude lower computational cost, thereby enabling the rapid evaluation of

materials over a large chemical space for promising candidates.

The accuracy of an ML model is significantly influenced by the selection of input features or representation. Because ML-based high-throughput screening tasks typically cannot include features from the DFT-optimized structure, tabulated atomic properties (such as radii) have been utilized as the inputs in the ML models used to screen materials generally,[9–11] and specifically for halide perovskites.[7,12–16] Some of us have previously reported a descriptor,[17] τ that allows for the classification of whether a compound will form a stable perovskite based only on its composition. Because τ and other composition-only models use only tabulated atomic information, it estimates stability very efficiently.

However, composition-based ML models have limitations in describing properties (e.g., the stability and band gaps), which can vary significantly depending on the atomic structure. This has been shown in the literature for perovskites specifically,[5,18–21] where local symmetry breaking resulting from BX_6 octahedral tilting and distortions significantly affects material properties. Therefore, models that incorporate structural information directly are preferred in order to account for this complication.

Numerous structure-based representations have been developed for materials, which aim to either quantify the local environment or catalog the bond distances and angles within the structure.[22–26] Alternatively, it is convenient to represent materials as a graph, where each atomic site is a node, and an edge corresponds to bonds between atoms (defined as two sites lying within a specified cutoff distance).[27,28] Numerical features can then be generated from the graph, such as through the counting of different paths through the graph, which can then be combined with some regression algorithm.[29] Alternatively, the graph representations can be combined with message passing neural networks, as first demonstrated for atomistic systems, to the best of our knowledge, in Ref.[27] These are well-suited for featurizing structure-based properties by directly processing the positions and distances between atoms as inputs into a neural network. This capability significantly reduces the necessity for manual feature engineering and typically achieves high accuracy in predicting various

properties of materials.[27,28,30–37]

However, a key challenge is generating training sets that include atomic structures without conducting a DFT calculation. This limitation is particularly significant in high-throughput screening for the discovery of currently *unknown* materials not tabulated in existing datasets. An alternative approach to bypass the need for DFT-optimized structures as input in the ML model involves several recently reported GNN-based ML interatomic potentials (MLIPs). Notable examples include M3GNet[38] and CHGNet,[39] as well as the MACE architecture.[40] All of these models were trained on snapshots of structures from DFT relaxations and thus, could be used to optimize the geometries of arbitrary new compounds.

An additional potential limitation of using GNNs is their requirement for a large number of samples to parameterize the model and achieve high accuracy. This presents a challenge when screening new materials, where the goal is often to explore a vast composition space with a minimal set of training samples. Hence, training a model with a scheme that allows for reusing information from other ML models trained to unrelated datasets would be highly beneficial.

To address the challenges posed by the availability of limited annotated data, an unsupervised domain adaptation (DA) approach has been adopted.[41] This previous work aimed to extract domain-invariant features from both the source and target domains. However, to the best of our knowledge, the DA approach has not yet been demonstrated for structure-based GNNs to predict material properties.

Here, we present a strategy that facilitates the application of GNNs to materials screening in three ways: 1) bypassing inputting the DFT-optimized structures by using an inexpensive and accurate proxy structure-generation scheme;[42] 2) improving the learning efficiency of GNNs via DA; and 3) enhancing model generalization to a large unlabeled space using active learning. We apply all three strategies to screen 222,960 AA'BB'X_6 halide perovskites based on three distinct targets that are relevant for the identification of 48 promising materials for PV applications: PBE-computed formation energies (ΔH_f^{PBE}) and band gaps (E_g^{PBE}),[43]

and HSE-computed band gaps (E_g^{HSE}).[44] To perform a comparative analysis, three representative GNN architectures were trained in this work; SchNet, which has permutational, translational, and rotational invariance;[27] PaiNN, which includes translational and permutational invariances, as well as, rotational invariances and covariances;[31] and ALIGNN, which incorporates both a line graph and a bond graph and has demonstrated strong performance across various benchmark datasets.[32]

Results and discussion

Dataset construction and preparation

A total dataset of 222,960 halide double perovskites was first generated by considering various combinations of 54 different elements occupying the A (A') and B (B') cation sites, and X = F, Cl, Br, and I. To ensure that the combination of elements charge balance, the cation oxidation states were assigned using the algorithm reported by Bartel et al.[17] A heatmap illustrating the frequency of each element's occurrence on the A (A') and B (B') sites within the 222,960 compositions is provided in Fig. 1. For all of the double perovskites, a rock-salt cation ordering (see Fig. 1A) within the cubic with $Fm\bar{3}m$ space group was assumed.

For all compositions, a cheap proxy structure was used as an input for the ML models, which was determined by minimizing the Global Instability Index (GII) following Ref.[45] These are referred to as BVM (bond valence method) structures throughout this work. These BVM structures compare well with DFT optimized structures with MAE = 0.39 $\mathring{A}$ for a set of 9096 compositions. The computational approach for generating the BVM structures is detailed in the Methods section. Note: The compositions evaluated in this work are restricted to the cubic symmetry which may result in an underestimation of stability and band gaps due to symmetry breaking. For example, a previous report showed that composition-specific symmetry-breaking distortions in a set of $Cs_2BB'Cl_6$ compounds resulted in an average increase in the band gap of 0.35 eV.[?]

The materials discovery workflow used in this work is as follows:

- Active learning and DA were combined to efficiently train separate ML models using three different architectures for the prediction of ΔH_f^{PBE}, $\Delta H_f^{ML,PBE}$, on a minimum set of training samples.
- $\Delta H_f^{ML,PBE}$ was used to determine the decomposition energies (ΔH_d^{PBE}) for each of the 222,960 AA'BB'X_6 compounds through the construction of a convex hull of all competing phases available in the OQMD database[46] to assess the stability relative to decomposition into other phases.
- A subset of 939 AA'BB'X_6 compounds were selected based on specific criteria, and separate ML models were trained to predict the HSE06 band gap energies, E_g^{HSE}. The training set for $E_g^{ML,HSE}$ was limited to a size that is practical because of the computational cost of generating a training set of E_g^{HSE} values is exorbitantly high.

In the third step of the workflow outlined above, the key criteria in selecting the materials is that they are stable based on $\Delta H_f^{ML,PBE}$. Additionally, we only considered compounds composed of chemically reasonable elements, excluding f-block elements such as lanthanide-containing compounds because of the computational expense and limitations of DFT in describing elements with f-electrons,[47] which is discussed in more detail below.

Domain adaptation

We evaluated the DA approach performance by comparing the model's performance using the mean absolute error (MAE) as a metric with the direct-training model. In the direct-training model, GNNs begin with randomly initialized weights, whereas in the DA approach, they start from pretrained weights, obtained from a pretrained model trained with the single perovskite dataset. For the DA approach, the choice of the source dataset was 500 ABX_3 compounds. We pretrained models using various source datasets and tested the models'

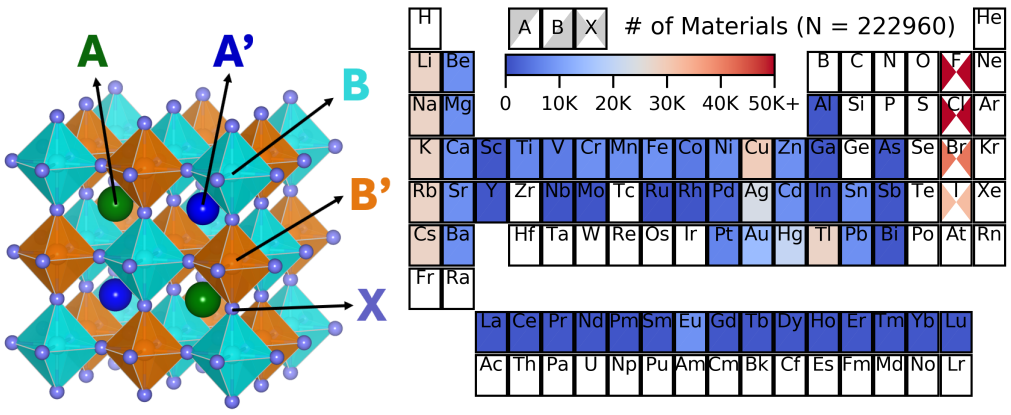


Figure 1: (a) Cubic double perovskite (AA'BB'X_6) conventional unit cell structure with rock-salt site ordering. The larger A and A' sites are surrounded by corner-sharing BX_6 and B'X_6 octahedra. (b) Periodic table chart showing the A-site (upper left corner), B-site (lower right corner), and X-site (left and right triangles) frequencies (heatmap) of each element present in our dataset of 222,960 halide AA'BB'X_6 compositions.

performance with the target dataset, AA'BB'X_6 compounds. We found that the 500 ABX_3 halides provides a considerably more accurate model than using the DA scheme with a source dataset of AA'BB'O_6 (i.e. oxides) compounds (see Fig. S.2). Thus, we can conclude that the source dataset that has similar chemical environment to that of the target dataset provides a better performance for the transfer learning.

We study the learning curve for both approaches by increasing the training set from 100 compounds up to 700 compounds while the test set is fixed to be the same 270 compounds for every model. Fig. 2 shows the MAEs of both DA and direct-training models, performed using three GNN architectures. The MAEs reduce drastically when the size of training set is larger. For SchNet and ALIGNN, this convergence occurred with 400 training samples, while for PaiNN, convergence was reached between 400 to 600 samples. These results indicate that the DA scheme effectively transfers knowledge of the chemical environments from the source dataset to the training of the target dataset. Although a substantial improvement is observed when the number of samples in the target dataset is increased, increasing the source dataset size had little effect on the retraining of the target model (see Fig. S.1).

Screening of unseen AA'BB'X_6 compounds using the ML-predicted formation energy

Rather than using the entire AA'BB'X_6 dataset which has 222,960 compounds for training which requires time-consuming DFT approach to create labels such as band gaps for all AA'BB'X_6, we select a small subset of samples for training. However, the training set constructed by choosing a small subset of samples can directly affect the accuracy of the ML model when encounter out of domain data. To address this, we used the active learning approach to improve the accuracy by strategically increasing the training set.

In Iteration 1 of active learning, we first randomly select subsets of AA'BB'X_6, which are 700 and 270 compounds, indicating as training set and initial test set (ITS), respectively. We integrate active learning with the DA approach by initializing three distinct pretrained models, each trained on a different split of single perovskites. Three pretrained models are then trained on the same 700 AA'BB'X_6 compounds. To test the performance, the models are used to predict properties of the remaining 221,990 AA'BB'X_6 compounds, assigned as an *unseen test set*. Each of the three models can predict different values of $\Delta H_f^{ML,PBE}$even applying to the same AA'BB'X_6 compound. Therefore, we computed the standard deviation for the prediction of each AA'BB'X_6 compound across three models $\sigma(\Delta H_f^{ML,PBE})$.

Fig. 3(a) presents violin plots for $\sigma(\Delta H_f^{ML,PBE})$ of Iterations 1, 2, and 3 for the SchNet:DA and PaiNN:DA models. SchNet:DA and PaiNN:DA indicate that we utilize SchNet and PaiNN by incorporating the DA approach. We set criteria for compounds with σ above the 99th percentile and below the 1st percentile to be high-uncertainty (HU) and low-uncertainty (LU) samples, respectively. We include the HU samples from the current iteration of active learning into the training set for the next iteration. In Iteration 2 (3), the number of training samples is hence 1970 (3593) by including 1270 (1673) HU samples from Iteration 1 (2). The models from the previous iteration are used as initialized weights for training models with a larger training set for the next iteration. (Is this correct??) We note that all Nd-containing samples are excluded from the HU samples due to their consistently high σ values (why??)

(see Fig. S.6) and inconsequential for PV applications.

Figure 3(b) illustrates progressions of the MAEs using the active learning and presents the MAES separated by the HU and LU samples. At Iteration 1 with 700 training samples, the direct approach yields MAEs higher than the DA approach when comparing the same GNN architecture. Therefore, we employ the DA approach for the active learning rather than the direct approach. At Iteration 3, the ALIGNN model demonstrates MAEs of 61 and 38 meV/atom for 790 HU and 948 LU samples, respectively. The MAE of LU samples is comparable to the ITS MAE of 37 meV/atom.

In contrast, the SchNet and PaiNN models yield MAEs for the HU samples 83 and 151 meV/atom, respectively, significantly higher than observed for ALIGNN. While their LU sample MAEs (48 meV/atom for SchNet and 47 meV/atom for PaiNN) exceed those of ALIGNN, they remain below thier respective ITS MAEs (51 meV/atom for SchNet and 61 meV/atom for PaiNN) (see Table S3). Therefore, the ALIGNN model was selected for screening the stable new compounds in the next section.

We emphasize that although both MAEs of the HU and LU samples significantly decrease driven by an increasing number of training samples, the MAE of ITS remains unchanged as the training sample size grows. This indicates that the improvement in HU and LU MAEs is not solely due to the number of training samples, but also the selective inclusion of samples in the training set through the active learning strategy. By utilizing the active learning, the MAEs of HU and LU samples decrease tremendously, and the LU sample MAE is similar to ITS MAE. Therefore, our method can maximize the accuracy of the ΔH_f^{PBE} prediction for unseen data while minimizing the number of the training samples.

Developing a machine learning model for predicting PBE and HSE band gap energies

As described in the section Dataset construction and preparation, we chose a subset of compounds that demonstrated stability according to the ML-predicted ΔH_f^{PBE} from Iteration 1

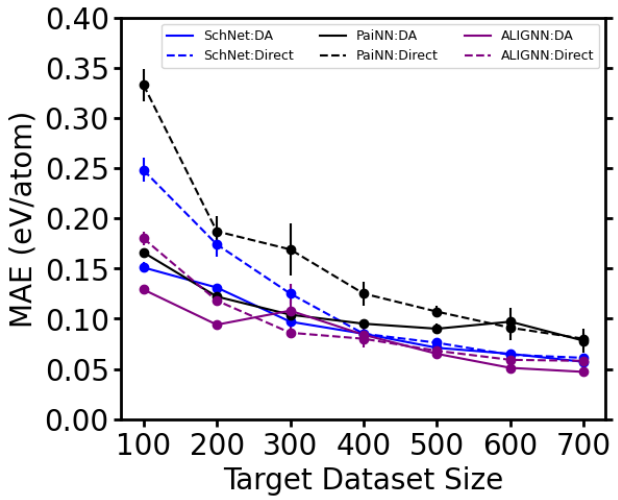
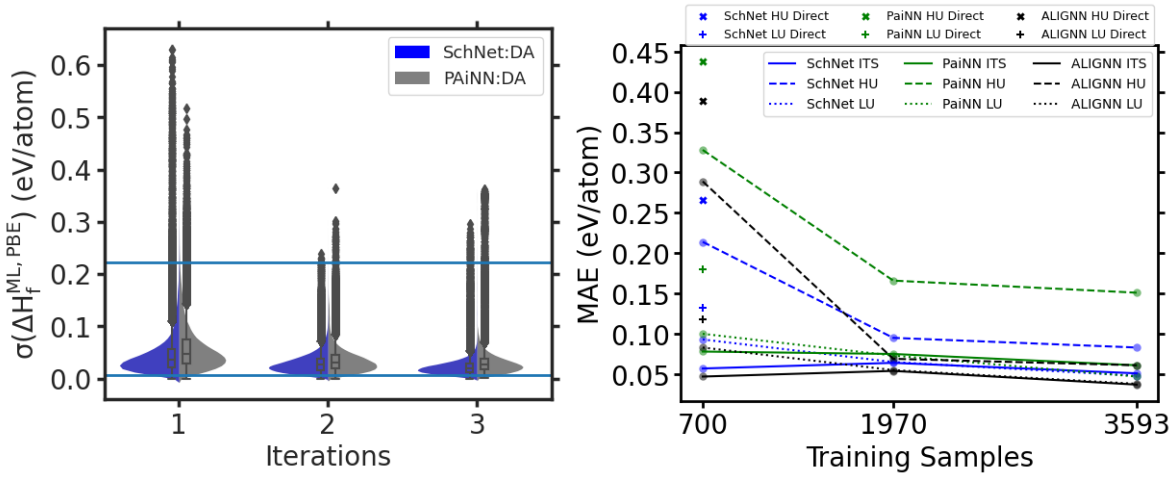
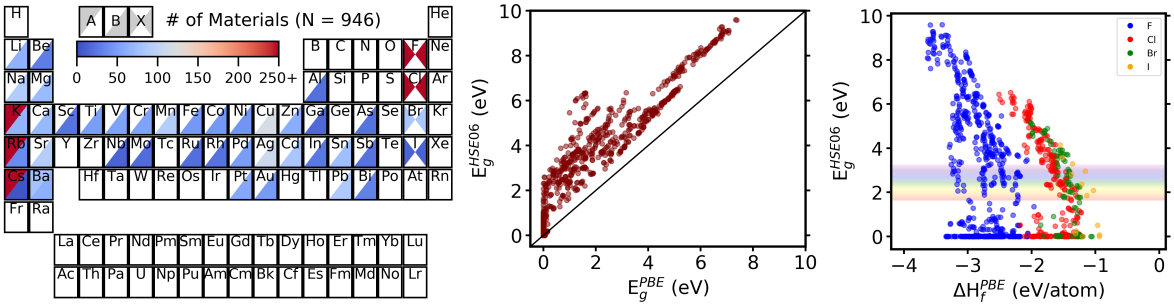


Figure 2: Change in the test-set MAE of $\Delta H_f^{ML,PBE}$ (compared with ΔH_f^{PBE}) calculated on a fixed test set of 270 samples, with the training set size increasing in increments of 100 samples. In all cases, 500 ABX_3 samples were used as the source dataset. Error bars indicate standard deviations over independent runs.



(a) (b)

Figure 3: (a) Distribution of the standard deviation across three independently trained $\Delta H_f^{ML,PBE}$ models over unseen samples, denoted $\sigma(\Delta H_f^{ML,PBE})$, using DA PaiNN and SchNet; (b) Evolution of MAE for predicted ΔH_f^{PBE}over three active learning iterations. The number of training samples in Iteration 1, 2 and 3 are 700, 1970, and 3593, respectively. The number of initial test-set (ITS), high uncertainty (HU), and low uncertainty (LU) samples in Iteration 1 are 270, 1679, and 1873, respectively. In Iteration 2, these numbers are 1873 (ITS), 1623 (HU), and 1877 (LU). In Iteration 3, the numbers are 1877 (ITS), 790 (HU), and 948 (LU), respectively.



(a) (b) (c)

Figure 4: (a) Periodic table plot showing the A-site (upper left corner), B-site (lower right corner), and X-site (left and right triangles) frequencies (heatmap) of each element present in our subset of 939 samples for which E_g^{HSE} was computed. (b) Parity plot showing that E_g^{PBE} consistently underestimates E_g^{HSE}. (c) Distribution of calculated E_g^{HSE} and ΔH_f^{PBE}.

for which E_g^{HSE} was calculated and $E_g^{ML,HSE}$was trained. The subset of 939 most frequently have K, Rb, Cs, and Ba on the A-site because of the higher likelihood for forming stable compounds when these elements occupy the A-site. Larger elements from Group 1 (Cs, Rb, K, Na) and Group 2 (Ba, Sr) will most likely occupy the A-site, which can be observed from the difference in $\Delta H_f^{ML,PBE}$ for compounds containing these cations on the A-site compared with the B site (Fig. S.5). In separate work, these elements have also been reported to enhance the stability of perovskites, partially because of their relatively large ionic radii.[47] Despite the training/test set being limited to only 939 compounds and predominantly consisting of compounds with only a few different elements on the A-site, the elements on the B and B' sites in this set span 35 different elements, despite excluding the lanthanides. This chemical diversity enables an investigation of how variations in the nature of the B-X bond affect the electronic properties. Fig. 4C compares the range of ΔH_f and E_g^{HSE}, with these compounds spanning over a 9 eV range in E_g^{HSE}. For this subset, the mean absolute error between E_g^{PBE} and E_g^{HSE} is 1.21 eV.

The most accurate models relative to E_g^{HSE}, are consistently based on the DA-scheme, which for SchNet and ALIGNN results in excellent accuracies of MAE = 181 meV and 188 meV using the BVM structure, with slightly higher errors are observed for PaiNN (236

meV). Slightly higher accuracies can be achieved when the PBE structure is used instead, but the small improvement in accuracy does not justify the considerable additional expense of performing a PBE optimization. Using the same training/test set split, the DA approach also leads to an improvement over the direct approach for E_g^{PBE} for SchNet (140 meV vs. 158 meV for DA and direct, respectively) PaiNN (122 meV vs. 172 meV) and ALIGNN (109 meV vs. 116 meV) for the excluded test set (see Table **??**).

Because of the computational cost of generating a source dataset of E_g^{HSE}, in addition to DA, transfer learning (TL),[48–50] was also used by leveraging an initial model was trained on E_g^{PBE} for 751 AA'BB'X_6 samples. In TL, a separate model was trained but initialized with the parameters from the E_g^{PBE}-trained model and fine-tuned for E_g^{HSE} (using the same 751 AA'BB'X_6 samples). The performance of these models is then assessed using the same 188 AA'BB'X_6 compounds. The TL approach resulted in MAEs for E_g^{HSE} of 190, 267 and 214 meV, for SchNet, PaiNN, and ALIGNN, respectively, on the test set of 188 compounds using the BVM structure as input. In comparison, the direct training approach led to MAEs of 245, 276, and 227 meV, respectively, indicating that the TL approach led to an improvement for all three models (Table **??**). Training the ML model to the PBE structure instead leads to similar results, indicating that the BVM structure, which is an inexpensive proxy structure, is a significant advantage in screening these materials.

We note that an alternative scheme for predicting E_g^{HSE} is Δ-learning, which was previously used by some of us to learn the HSE06 quality band structures.[51] However, Δ-learning requires inputting E_g^{PBE} to predict E_g^{HSE} – i.e., requires a successful PBE geometry optimization prior to making a prediction – which is more computationally demanding than the DA/TL schemes that make it possible to predict E_g^{HSE} directly from the BVM structure.

Separately, the subset of 939 compounds were selected from Iteration 1 and not used in subsequent active learning iterations, and therefore, represent an additional totally left-out set for evaluating the performance of ML models from later iterations in predicting ΔH_f^{PBE} to assess the generalizability of the ML models. This subset shows a different distribution

of ΔH_f^{PBE} compared to training samples in Iteration 1 (see Fig. S.4) indicating that they represent an out-of-domain distribution. As a result, the DA-based models trained in Iteration 1 had high errors for ΔH_f^{PBE} for these samples specifically (MAE = 107, 159, and 142 meV/atom for SchNet, PaiNN, and ALIGNN, respectively) (see Table S.2). However, these errors were factors of 2.0, 1.9, and 1.7 lower than the directly-trained ML models from Iteration 1 (MAE = 215, 208, and 244 meV/atom, respectively). The higher accuracies for the DA-approach indicate an enhancement in the ML model's generalizability when trained through the DA approach. By Iteration 3, the accuracies of the DA-models improved significantly to MAE = 45, 56, and 34 meV/atom for SchNet, PaiNN, and ALIGNN, respectively (as shown in Table S.2). This trend aligns well with the errors observed for the excluded HU and LU samples, which also showed that sufficient accuracy in $\Delta H_f^{ML,PBE}$ could be obtained by Iteration 3.

Screening materials based on $\Delta H_f^{ML,PBE}$ and $E_g^{ML,HSE}$

The DA ALIGNN models trained to predict $\Delta H_f^{ML,PBE}$ (description in Section Screening of unseen AA'BB'X_6 compounds using the ML-predicted formation energy) and $E_g^{ML,HSE}$ (description in Section Developing a machine learning model for predicting PBE and HSE band gap energies) for an unseen set of 213,200 compositions as well as adding the compounds from the training set, leading to evaluation of properties for 222,960 samples in Fig. 5. Note that these predictions *include* compounds containing f-block elements, even though DFT struggles with these elements and they were intentionally excluded from the training set for $E_g^{ML,HSE}$. Therefore, in these cases, the predictions are expected to be unreliable, yet they were still included (but are labeled as such in Table S.6).

The $\Delta H_f^{ML,PBE}$ values are negative indicating stability against the competing elemental phases, except for a few cases that generally contain X = I. Compounds that contain X = F tend to exhibit the lowest $\Delta H_f^{ML,PBE}$ values followed by X = Cl, Br, and I. $\Delta H_f^{ML,PBE}$ is used to compute $\Delta H_d^{ML,PBE}$through the convex hull construction using all competing phases

available in OQMD (see Methods section for more details). To determine the criteria for stability, we selected 46 experimentally reported compounds in the ICSD that are not in the training set for our ML model. DFT-calculated Δ H_d indicates all of these 46 compounds have an Δ H_d ¡ 70 meV/atom for all compounds. In comparison, the ML-predicted Δ H_d has a high accuracy relative to DFT of 56 meV/atom (see SI Table X). The signed error between the ML-predicted Δ H_d and DFT is -50 meV/atom, indicating that the ML model will predict compounds as unstable when they are stable, which should prevent false positives. Therefore, a large cutoff enthalpy of 60 meV/atom was chosen to be +20 meV/atom larger than the MAE of the ML model (38 meV/atom) to ensure candidate materials that may be under stabilized by our model are not artificially excluded. Out of the 222,960 compositions, only $\approx$ 2 % (3410 samples) are predicted to be stable/meta-stable with $\Delta H_d^{ML,PBE} < 60$ meV/atom.

Regarding band gaps, we selected a range of 1.0 eV $< E_g^{ML,HSE} <$ 1.6 eV for the screening process. This range accounts for the Shockley–Queisser limit of 1.34 eV, but expanded by $\pm$ 0.3 eV because of the error in the $E_g^{ML,HSE}$. Of the 222,960 compounds evaluated (i.e., all training and test set samples were considered in this screening), only 48 candidate compounds are within the range of 1.0 eV $< E_g^{ML,HSE} <$ 1.6 eV and are relatively stable with a $\Delta H_d^{ML,PBE} < 60$ meV/atom. Note: this set excludes 69 f-block compounds that fall within this range but are considered unreliable. Moreover, an additional 21 compounds with Tl on the A-site were excluded because of the toxicity of Tl.

These 48 candidates are a chemically diverse set, containing a total of 6 elements on the A-sites and 17 elements on the B-sites. The majority of candidates predominantly have 3 elements on the A-sites (Cs, Rb, K) and 3 elements on the B-sites (Cu, In, Rh). 67 % of the 48 candidate materials contain either Cu, In, or Rh on the B-site. Regarding the X-site, the range and magnitude of $E_g^{ML,HSE}$ increases as the radius of X decreases from I to Br, Cl and F. This agrees with previous computational results, where stability and E_g tends to increase as X becomes smaller (from I to F).[13,47] Because of the general trend towards increasing

band gap energies in substituting anions from I to F, the set of 48 candidates contains Cl (50 %) most frequently, followed by Br (27 %), and then F (23 %).

To verify the accuracy of the model, we randomly selected 8 compounds from the set of 48 and subsequently calculated E_g^{HSE} using the PBE optimized geometry. These 8 compounds were chosen such that B/B' = Cu, Sb, In, Rh, and/or Ag because these elements are the 5 most frequently observed B/B'-sites in the candidates predicted with and without the DA approach. These errors are reported for each of the 8 compounds in Table S.5 and show an MAE = 0.14 eV. The projected density of states for 6 of these 8 compounds is displayed in Fig S.7. Moreover, several of the 48 candidates have been previously investigated for their potential in PV applications. These prior studies provide a useful basis for comparison when assessing the accuracy of $E_g^{ML,HSE}$ against independent results. Included in this set of 48 candidates are five compounds present in the training data set ($RbCsAgInBr_6$, $Cs_2GaAgBr_6$, $Cs_2AgInBr_6$, $KRbCuRhF_6$, $KCsCuInCl_6$), two of which have been already investigated, $Cs_2GaAgBr_6$[52] and $Cs_2AgInBr_6$,[52] and as expected compare well with $E_g^{ML,HSE}$(see Table 1). Six compounds in the test set have been previously investigated: $Cs_2CuSbCl_6$,[53] $Rb_2CuInCl_6$,[52] $Rb_2AgInBr_6$,[52] $K_2CuGaCl_6$,[52] $K_2CuInCl_6$,[52] and $Rb_2GaAgBr_6$,[52] and their reported E_g^{HSE} values are on average signed difference of -0.14 eV (indicating consistently underestimating other calculated results) with the largest difference of -0.31 eV of $E_g^{ML,HSE}$ for $Rb_2CuInCl_6$. Finally, four of the 48 candidate materials identified here were previously disregarded. $Cs_2CuInCl_6$, $Cs_2GaAgBr_6$, $K_2CuInCl_6$, and $K_2CuGaCl_6$ failed the criteria for stability with respect to binary competing phases.[52] Overall, the strong agreement between these ML predictions and the independently reported values for these compounds — 4 of which have been reported in the literature as being stable — further emphasizes the potential of the 48 candidate compounds identified in this study.

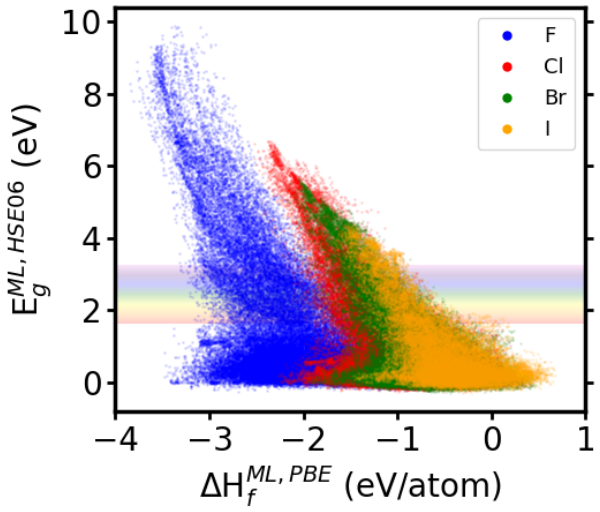


Figure 5: The DA ALIGNN model $\Delta H_f^{ML,PBE}$ and $E_g^{ML,HSE}$ for 222,960 perovskite compositions. Compounds with X = I tend to possess higher ΔH_f and smaller E_g whereas compounds with X = F tend to have the lowest $\Delta H_f^{ML,PBE}$ and largest spread of E_g. Compounds with X = Br, Cl tend to have intermediate ΔH_f and E_g with Cl possessing the larger E_g on average.

Table 1: Comparison of the $E_g^{ML,HSE}$ for samples selected from the set of 939 with previously reported E_g^{HSE} values. All values are in eV

Compound	**Dataset**	**Exp.** E_g**(eV)**	E_g^{HSE}**(eV)**	$E_g^{ML1,HSE}$**(eV)**	$E_g^{ML2,HSE}$**(eV)**	$E_g^{ML3,HSE}$**(eV)**
$Cs_2GaAgBr_6$	Test	-	1.32	1.02	1.03	1.07
$Cs_2AgInBr_6$	Training	1.56	1.25-1.50	1.31	1.31	1.32
$Cs_2CuSbCl_6$	Test	1.66	1.70	1.82	1.76	1.57
$Rb_2CuInCl_6$	Test	-	1.36	1.00	1.10	0.96
$Rb_2AgInBr_6$	Test	-	1.46	1.41	1.28	1.40
$K_2CuGaCl_6$	Test	-	1.30	1.57	1.29	0.99
$Cs_2CuInCl_6$	Test	-	1.40	1.25	1.24	1.09
$K_2CuInCl_6$	Test	-	1.35	0.96	0.97	0.97

Methods

To create a system-specific structure for each of the 222,960 compounds, the optimal lattice parameters for the 10-atom unit cell were obtained by minimizing the Global Instability Index (GII) for each composition. GII quantifies the deviation from ideal bond valences and serves as an indicator of structural stability. To calculate bond valences, oxidation state-specific cation-anion bond valence parameters from the 2020 update by Brown et al.,[54] and only cation-anion bonds less than 6 Å were considered. The minimization of GII for structure

generation has previously been shown to accurately reproduce experimental[55] and theoretical[42] perovskite structures on a large scale.[45] The structure prediction code is publicly available at: https://github.com/zaba1157/bvm_cubic_perovskite. The same GII minimization scheme was used to generate the primitive structures for all 500 single perovskites with the formula ABX_3, all within the cubic with the $Pm3m$ space group (#221).

Band gap energies were computed using a fully converged single self-consistent field calculation on the PBE-optimized geometries with both the PBE and HSE06 functionals (simply referred to as HSE). The HSE calculations are performed without consideration of spin-orbit coupling, and the standard 25% exact Fock exchange is included. The tetrahedron method with Blöchl corrections and a Γ-centered k-mesh were used for all band gap calculations. All DFT calculations were performed within the PAW formalism as implemented in the Vienna ab initio simulation package (VASP) version 5.4.4.[56–58] These PBE geometry optimizations were performed with a plane-wave cutoff energy of 600 eV and an energy convergence criterion of 10^{-6} eV. The relaxation was stopped when the change of the total energy was smaller than 1E-5 between two ionic steps. The tetrahedron method with Blöchl corrections and a Γ-centered k-mesh were used for all band gap calculations. A Gaussian thermal smearing with a parameter k_BT of 0.02 eV was used in all geometry optimizations. $\Gamma-$centered k-meshes were used with a grid spacing of less than 0.1 $Å^{-1}$ for the Brillouin zone sampling. All magnetic systems were initialized in a ferromagnetic spin state. Initial magnetic moments of 5 μ_B was used for d-block elements: Sc, Y, Ti, V, Nb, Cr, Mo, Mn, Fe, Ru, Co, Ni, and Pt, and 7 μ_B as used for f-block elements: Ce, Pr, Nd, Pm Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, and Lu.

Formation enthalpy is calculated by $\Delta H_f = \sum_{i=1}^{N} \mu_i * n_i$ where n_i is the number of atoms of element i and μ_i is elemental chemical potential. The elemental chemical potentials are taken as the "fit-partial" reference states reported in the OQMD for compatibility.

SchNet[27] is a continuous-filter convolution neural network that encodes information from the atomic local environments by first determining the neighboring atoms within an 8 Å cut-

off distance. Then, to encode many-body interactions from these local environments, message passing is performed from three key components: atom-centric layers, interaction modules, and filter generation networks.

The atom-centric layers perform transformations on atomic embeddings, derived from atomic numbers, using multiple dense layers with each layer having a width of 128. SchNet's convolution module updates edge features through localized pooling, integrating the information from neighboring nodes. The interaction module further combines node and edge features through three dense layers and an addition module, with each layer again having a width of 128 and employing a shifted softplus function for non-linear activation. Downstream predictive tasks leverage a multi-layer perceptron with several dense and pooling layers. The network's depth was consistently set to eight for all experiments. The Keras Graph Convolution Neural Networks (KGCNN)[59] package has been used in all the experiments.

PaiNN is an equivariant neural network that captures the direction of interatomic distances using directional vectors, in contrast to the scalar representation of distances used by models such as SchNet. It transforms the Euclidean distances between atom pairs into an equivariant representation using a Bessel basis with 20 radial functions, optimized for a 3 Å cutoff. We tested larger cutoffs of up to 6 Å without significant performance gains. Each radial function passes through a series of layers—input, hidden, and output—to calculate atomic interaction weights. For convolution, PaiNN applies a sequence of three dense layers, paired with two pooling and two gathering layers to integrate invariant and equivariant features—a process executed via the Keras library.[59] This is followed by an update block that refines both node and equivariant features, utilizing four dense layers along with various geometric and arithmetic modules. Like SchNet, PaiNN has been tested in the KGCNN[59] environment.

ALIGNN combines a bond graph from interatomic distances with a line graph.[32] The nodes of the bond graphs represent the atoms and the bonds between atoms are represented with edges while the nodes of a line graph refer to interatomic bonds and its edges indicate

the bond angles. This kind of representation has facilitated the inclusion of more information from the structures which eventually ensures better learning and so the prediction of different properties.[32] The edge gated graph convolution of ALIGNN update node features and edge features concurrently. The node feature set is the combination of 9 inputs, which is inspired by the crystal graph convolutional neural network (CGCNN) model.[30] The initial edge features are composed of interatomic bond distances expanded via radial basis functions. Each ALIGNN layer consists of edge-gated graph convolution over the line graph followed by another edge-gated graph convolution over the bond graph. The first convolution produces the bond messages that are used by the next edge-gated graph convolution module to update the node feature and bond features of the original graph. The convolution had been performed using 4 ALIGNN layers and then performed edge-gated graph convolution for 4 layers afterward with 256 hidden dimensions and the sigmoid linear unit (SiLU) for activation function. After graph convolution, the graph-level features were obtained using the *mean global pooling* across all nodes before passing through fully connected layers to obtain target values.

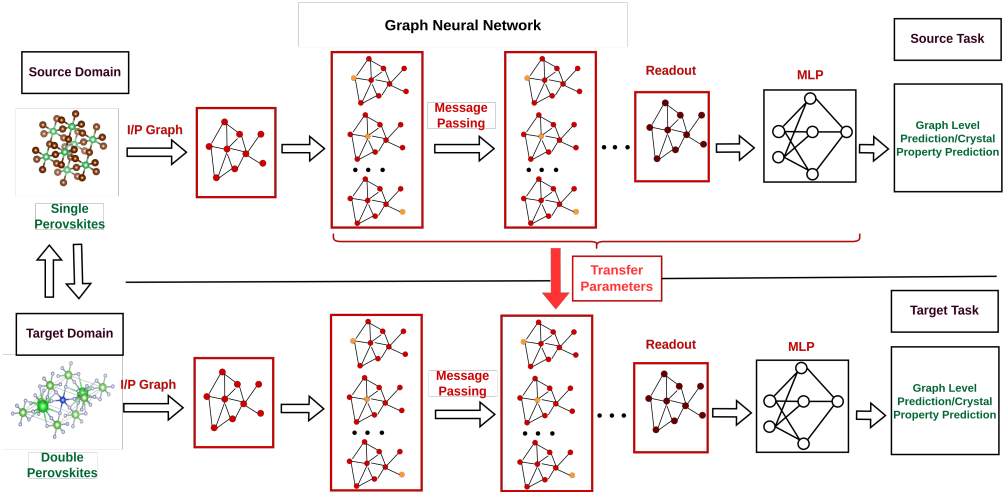


Figure 6: The DA approach involves two distinct pipelines, each with a different dataset: source and target. For the source dataset, 500 ABX3 compounds were used to train an initial model. In the DA scheme, the optimal parameters/weights from the source dataset training are transferred to begin training a new ML model on the target dataset. This model is then further refined through standard validation and testing procedures. The final target model is used to predict the properties of a new set of 222,960 AA'BB'X6 compounds.

Domain Adaptation

The DA method is to exploit the knowledge the model learned from one domain, a *source domain*, and use to improve the model's performance for other related *target domains*. There are three DA strategies, used for different objectives:[60,61]

- **Feature-Based DA** reduce the divergence of the latent features between source and target distributions.
- **Instance-Based DA** re-weights training data to correct the difference between source and target distributions.
- **Parameter-Based DA** passes parameter/weights from the pre-trained model to the target model to build a suited model for the target task.

Because generating a DFT-calculated training dataset for compounds with a large number of atoms is computationally demanding, our goal is to recycle the learnable parameters and weights that is learned in the source training and apply it to new systems. To do this, we used the parameter-based DA approach. Feature-based DA requires a new encoding mechanism or a labeling function for the target domain while instance selection and instance weighting is followed by the instance based DA to learn the importance of labeled data.[62] However, due to having limited source and target data, we decided to use the data unmodified. Therefore, we worked on the model level specifically with the weights and parameters rather than the dataset to deploy DA. The parameter-based DA approach, which is illustrated in Fig. 6, first involves training an ML model on an initial dataset (called the source dataset), then the parameters of this initial model are used to initialize a subsequent ML model for training to the target dataset. Our specific application of GNN architectures of materials involves transferring weights and parameters of different layers learned on the source dataset to a target dataset. The GNNs are the combination of message passing layers and MLP layers. In the transfer process, we transfer the weights and parameters for both of the message passing layers and MLP layers.

In the domain adaptation, the learning rate between source and target model training plays an important role. We have observed the differences in performance while trying different combination of learning rate between source and target model. We have tried to find an optimal learning rate for the domain adaptation scheme which can outform the best performing model from direct training. Learning rate optimization aims to find the best learning rate for domain adaptation(DA) and direct training protocol. After finding the optimal learning rate for both of the protocols we have compared the performance of DA and direct aprroach for their best learning rates. The results have been demonstrated in the following tables(next page). We have tested these experiments for two target properties: formation energy ΔH_f and E_g^{HSE}. In this process, we have completed three iterations of active learning. After the first iteration, we have only performed the DA as this approach showed better performance for initial test set(ITS) and generalization compared to the direct approach. The generalization scheme has been tested over the low uncertain(LU) and high uncertain(HU) samples queried from the 222960 unlabeled sample space based on their uncertainty. LU1 and HU1 indicates the samples selected for the first iteration, LU2 and HU2 for the second iteraton, and the LU3 and HU3 for the third iteration. From the second iteration, we only tried two combination of source LR and target LR which showed the better performance in the first iteration. Here, all of the results have been reported for ALiGNN. After all the experiments, it's evident that starting the source training with higher learning rate(0.001) and continuing the target training with lower learning rate(0.0005) has the advantage over most of the DA and direct experiments. Basically, the higher learning rate helps the source training to reach the convergence faster avoiding the overfitting while the lower learning rate for the target training preserves the useful knowledge while learned during pretraining preventing the forgetting issues.

The source ML model was trained on the source dataset for a total of 250 epochs, with 32/50 samples per batch. At each step within an epoch, the error of the model predictions were calculated against the actual labels for these 32/50 samples. Using different batch size,

doesn't make much impact over the performance. The optimal parameters and weights from the initial training were used to initialize the retraining on the target dataset. Different learning rate was used across all training of the source and target models and the MAE was used as the evaluation metric. An end-to-end training of the source dataset and retraining of the target dataset was performed three times, yielding three distinct models.

In case of checkpoint selection for applying the model over unseen test set and generalization, we have tried with both of the best checkpoint and last checkpoint. The last checkpoint shows more stability in the performance.

Conclusions

In this work, we demonstrate that domain adaptation (DA) improves the performance of ML models consistently across three different architectures applied to three different target properties of ΔH_f^{PBE}, E_g^{PBE}, and E_g^{HSE}. DA allows existing models to be efficiently fine-tuned for new domains or tasks, significantly reducing the resources and data typically required for training robust ML models from scratch. Indeed, combining the DA scheme with active learning, high accuracies were achieved in the prediction of ΔH_f^{PBE} (MAE = 38 meV/atom) andE_g^{HSE} (MAE = 0.18 eV) using training sets with a relatively small number of compounds. We also observed that the DA scheme improved the generalization of the ML models. To our knowledge, these results are the first demonstration of the DA scheme for GNNs to predict material properties.

Moreover, these accurate models allowed for screening a large space. First, $\Delta H_f^{ML,PBE}$ was used to compute the decomposition enthalpy ($\Delta H_d^{ML,PBE}$) through convex hull construction for all 222,960 compounds, which utilized all competing phases available in OQMD.[46] Of the 222,960 compositions analyzed based on $\Delta H_d^{ML,PBE}$, only ≈ 2 % (3410 compositions) are predicted to be (meta)stable with $\Delta H_d^{ML,PBE} < 60$ meV/atom. This set was further reduced when considering the band gap energies of the material. Of the 222,960 compounds,

only 48 candidate compounds fall within the desired range of 1.0 eV $< E_g^{ML,HSE} <$ 1.6 eV and display $\Delta H_d^{ML,PBE} <$ 60 meV/atom. Four of these were previously disregarded due to instability and 4 have been previously identified as promising for PV applications, which aligns well with our ML model predictions. The remaining dataset of 40 compounds remains unexplored to the best of our knowledge and therefore offer opportunities for potential experimental examination.

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Supporting Information Available

This will usually read something like: "Experimental procedures and characterization data for all new compounds. The class will automatically add a sentence pointing to the information on-line:

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